

**Problem 1 (20 points):** Solve for all possible values of the real numbers  $x$  and  $y$  in the following equations

$$(x + iy)^3 = -1$$

$$\therefore (x + iy)^3 = r^3 \cdot e^{3i\theta} = -1 = e^{i(\pi+2k\pi)}$$

$$\therefore r = 1 \quad \theta = \frac{\pi+2k\pi}{3}$$

$$\therefore x = \cos\left(\frac{\pi+2k\pi}{3}\right) \quad y = \sin\left(\frac{\pi+2k\pi}{3}\right)$$

$$\therefore x = -1 \quad y = 0 \quad \& \quad x = \frac{1}{2} \quad y = \pm \frac{\sqrt{3}}{2}$$

or:

$$\therefore (x + iy)^3 = -1$$

$$\therefore (x^3 - 3xy^3) + i(3x^2y - y^3) = -1$$

$$\therefore \begin{cases} (x^3 - 3xy^3) = -1 \\ (3x^2y - y^3) = y(3x^2 - y^2) = 0 \end{cases}$$

$$\therefore x = -1 \quad y = 0 \quad \& \quad x = \frac{1}{2} \quad y = \pm \frac{\sqrt{3}}{2}$$

**Problem 2 (20 points):** Evaluate each of the following in  $x + iy$  form

$$i^{2i/\pi}$$

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$$= e^{\frac{2}{\pi} \cdot i \ln i}$$

$$= e^{\frac{2}{\pi} \cdot i \cdot i(2k\pi + \frac{\pi}{2})}$$

$$= e^{-4k-1}$$

**Problem 3 (25 points):** Find the real and imaginary parts of the following function and determine if the function is analytic:

$$f(z) = \cos \bar{z}$$

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$$= \cos(x - iy)$$

$$= \cos x \cos(iy) + \sin x \sin(iy)$$

$$= \cos(x) \cdot \frac{e^y + e^{-y}}{2} + i \sin(x) \cdot \frac{e^y - e^{-y}}{2}$$

$$\therefore \frac{\partial u}{\partial x} = -\sin(x) \cdot \frac{e^y + e^{-y}}{2}$$

$$\frac{\partial u}{\partial y} = \sin(x) \cdot \frac{e^y + e^{-y}}{2}$$

$$\therefore \frac{\partial u}{\partial x} \neq \frac{\partial u}{\partial y}$$

$\therefore$  The function is not analytic.

**Problem 4 (30 points):** Find the real and imaginary parts of the following function and determine if the function is analytic:

$$f(z) = \cos \bar{z}$$

and show that

$$\overline{\cos \bar{z}} = \cos z$$

Is the following relationship true?

$$\overline{\sin z} = \sin \bar{z} ?$$

同作业 Problem 9 问题以及答案

Problem 9

$$\begin{aligned} \text{(a)} \quad \overline{\cos \bar{z}} &= \frac{\overline{e^{i\bar{z}} + e^{-i\bar{z}}}}{2} \\ &= \frac{\overline{e^{i\bar{z}} + e^{-i\bar{z}}}}{2} \\ &= \frac{e^{-i\bar{z}} + e^{i\bar{z}}}{2} \\ &= \cos z \end{aligned}$$

$$\begin{aligned} &f, g \text{ are complex numbers} \\ &z = f + ig \Rightarrow \bar{z} = \bar{f} - i\bar{g} \\ &z = \frac{f}{g} \Rightarrow \bar{z} = \frac{\bar{f}}{\bar{g}} \\ &\overline{e^{iz}} = e^{-i\bar{z}} \end{aligned}$$

$$\text{(b)} \quad \overline{\sin \bar{z}} = \frac{\overline{e^{i\bar{z}} - e^{-i\bar{z}}}}{2i} = \frac{e^{-i\bar{z}} - e^{i\bar{z}}}{-2i} = \sin z$$

**Problem 5 (20 points)** Show that

$$|e^z|^2 = e^{2\operatorname{Re}(z)}$$

设  $z = x + iy$

$$e^z = e^x(\cos y + i \sin y)$$

$$|e^z| = e^x \sqrt{\cos^2 y + \sin^2 y} = e^x$$

$$|e^z|^2 = e^{2x}$$

$$e^{2\operatorname{Re}(z)} = e^{2x}$$

得证

或者直接

$$|e^z|^2 = |e^x \cdot e^{iy}|^2$$

$$|e^{iy}| = 1$$